機械所碩士班學生研究成果

利用實驗與模擬監測體積成形機械與其產品品質的方法

Methodologies of Monitoring Bulk-Forming Machinery and Quality of its Products through Experiment and Simulation

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Summary

The manufacturing of fasteners with diverse shapes and sizes entitled with diverse specification requires credible and reliable bulk-forming machinery. In this research, both experimental data and simulation techniques are used to analyze the general behavior of the machinery that operates with or without fabricating parts. The results of the analysis are used to establish the monitoring methods that detect anomalies in the machinery and the products for building reliable and credible machineries. The methodology is firstly to equip the machinery with sensors. The data acquired are the A/D-converted signals from an accelerometer and a positioning mechanism, which consists of a set of rack and pinion and an optical encoder. Secondly, with the help of kinematics and the positioning signals, the voltage signals of acceleration were processed to obtain the horizontal components of acceleration, velocity, and displacement of the sliding block mechanism (SBM) with respect to time. Thirdly, the time series of acceleration and displacement were used to generate (1) vibration RMS value, (2) vibration spectrum, and (3) the power of low-frequency vibration with respect to displacement for each cycle. The time series of velocity and displacement were used to produce (1) the specific kinetic energy absorption of the SBM and (2) the horizontal force exerted by the SBM with respect to displacement. Fourthly, the resultant graphs and curves were compared with those resulted from simulation with Qform® software or with each other of normal and abnormal situations. The last step of the methodology is to use all kinds of comparative analysis techniques, such as machine learning, for the diagnosis or prognosis of the anomalies of the machinery or the manufactured products with the minimized additional equipment's and costs.

Keywords: Cold forging, Vibration analysis, Diagnosis of product anomalies, Prognosis of machinery failure.

Methodology

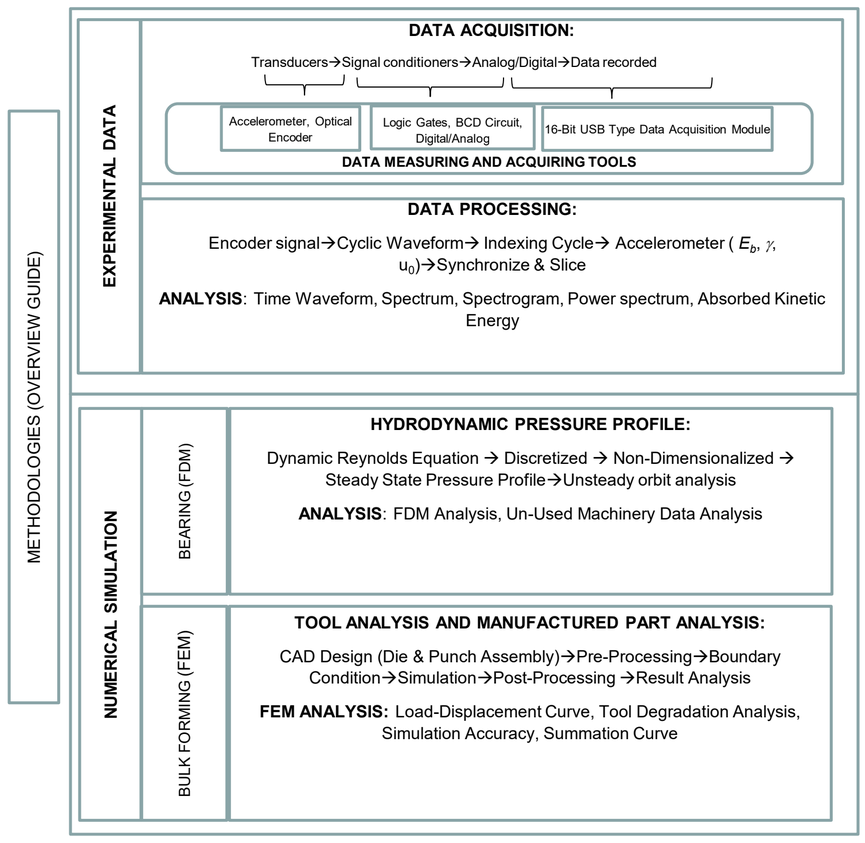


Fig.1 Methodology chart

Experimental Setup

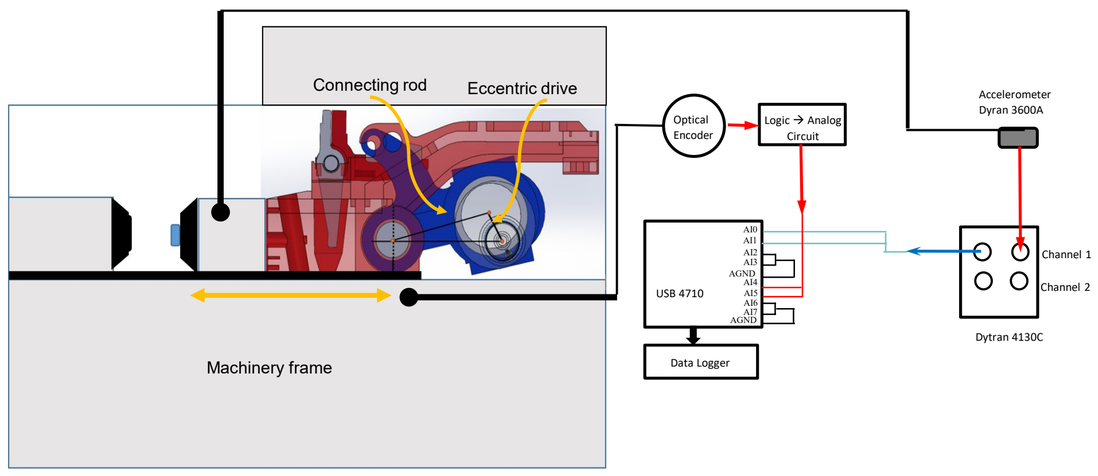


Fig. 2 Schematic representation of machinery (Left) and Data Acquisition System (Right).

Results and discussions

Simulation Summation Load Curve

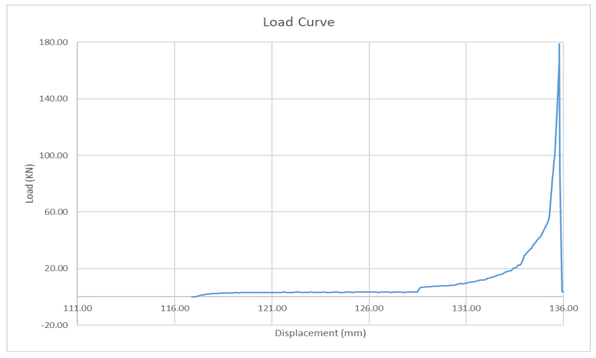


Fig. 3 Load curve of plus head screw

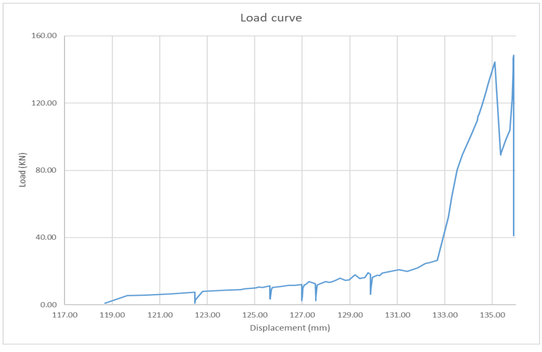


Fig. 4 Load curve of ASSY head screw

Experimental Data Analysis

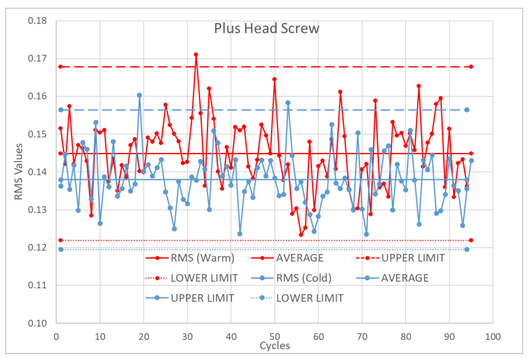


Figure 3-23 RMS value for 95 cycles of plus head screw warm and cold.

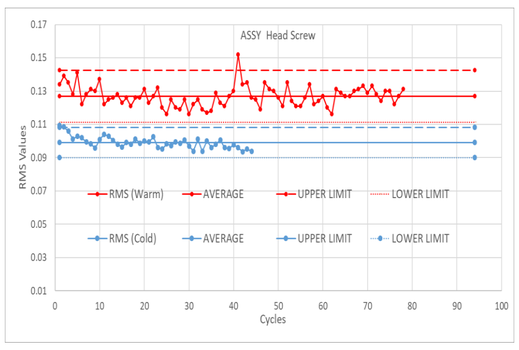


Figure 3-25 RMS value for 78 cycles of ASSY warm and cold cycles.

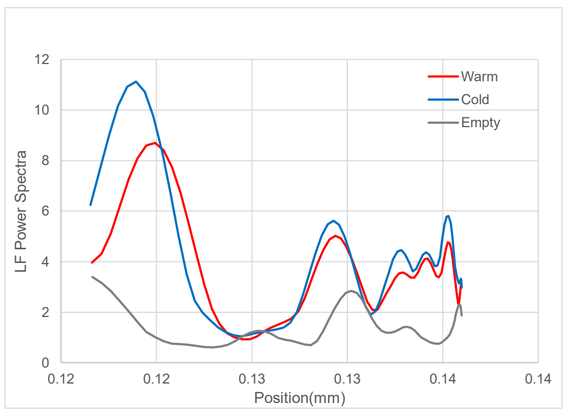


Figure 3-36 Power spectrum of forging stroke for plus screw head.

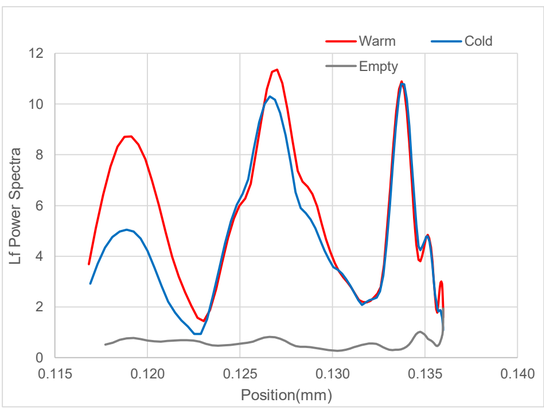


Figure 3-38 Power Spectrum of forging stroke for ASSY head.

Conclusions

The parameters, trends and patterns resulting from the monitoring methods has been analyzed. The analysis leads to the following conclusions:

1. The RMS and mean vibrational value of each cycle follows a trend when analyzed for warm and cold condition of the machinery.
2. The spectrum and spectral plot of the vibrational data shows the forging feature that could be used for the tracking and tracing of the ongoing variation of the forging signals.
3. The low-frequency power spectrum obtained by short time Fourier transform (STFT) of acceleration varying with time shows obvious peaks associated with each forging stations.
4. The absorbed specific kinetic energy and force of the sliding block computed by velocity-displacement plot of defined cycles shows good recognizing features for empty, warm, and cold condition machinery.
5. The simulated load curves when compared with various experimental data curves shows similar featuring anomalies that could be considered as the monitoring parameters.

With the methods and the results proving the feature extraction capability the discussed methods in future could be used for training various deep learning models with channelized algorithms for condition monitoring as well as prognosis.